

Challenge (Habanero)

- Q1.** Solve $|x| = 5$.
(A) $x = 5$
(B) $x = -5$
(C) $x = 5$ or $x = -5$
(D) $x = 0$
(E) $x = 10$
- Q2.** A recipe requires $|2x - 3| = 7$ cups of flour. Solve for x .
(A) $x = 2$ or $x = 5$
(B) $x = -2$ or $x = -5$
(C) $x = 2$ or $x = -5$
(D) $x = 5$ or $x = -2$
(E) $x = 0$ or $x = 10$
- Q3.** Solve $|3x + 2| = 11$.
(A) $x = 3$ or $x = -5$
(B) $x = -3$ or $x = 5$
(C) $x = 2$ or $x = -4$
(D) $x = 1$ or $x = -3$
(E) $x = 0$ or $x = 6$
- Q4.** A building has $|4x - 5| = 15$ meters of foundation. Solve for x .
(A) $x = 4$ or $x = 2.5$
(B) $x = -4$ or $x = -2.5$
(C) $x = 5$ or $x = 0$
(D) $x = 2$ or $x = -3$
(E) $x = 1$ or $x = -4$
- Q5.** Solve $|2x| = 12$.
(A) $x = 6$
(B) $x = -6$
(C) $x = 6$ or $x = -6$
(D) $x = 0$
(E) $x = 12$
- Q6.** A contractor needs $|x + 4| = 9$ blocks of concrete. Solve for x .
(A) $x = 5$ or $x = -13$
(B) $x = -5$ or $x = 13$
(C) $x = 3$ or $x = -11$
(D) $x = 1$ or $x = -7$
(E) $x = 0$ or $x = 8$
- Q7.** Solve $|5x - 3| = 2$.
(A) $x = 1$ or $x = 0.2$
(B) $x = -1$ or $x = -0.2$
(C) $x = 0.6$ or $x = 0.2$
(D) $x = 0.4$ or $x = -0.2$
(E) $x = 0.2$ or $x = -0.6$
- Q8.** A chef has $|3x - 1| = 8$ kilograms of ingredients. Solve for x .
(A) $x = 3$ or $x = -2$
(B) $x = -3$ or $x = 2$
(C) $x = 2$ or $x = -1$
(D) $x = 1$ or $x = -3$
(E) $x = 0$ or $x = 9$

Open Ended Questions (FRQ)

- Q9.** Solve $|2x + 5| = |x - 2|$ and check for extraneous solutions.
- Q10.** A construction team needs to measure $|x - 3| = |2x + 1|$ feet of material. Solve for x and explain your reasoning.

Chef's Tips

- Remember to check for extraneous solutions.
- Use the definition of absolute value to justify your answers.
- Ensure students provide clear explanations for their reasoning.